# Lab Assignment 7: Multiobjective Optimization

### 37 Algorithms, Trade-offs, Empirical Benchmarks and Applications

**Optimization Techniques Laboratory · M.Tech in Artificial Intelligence & Data Science**

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**Roll Number Suffix:** 13 (Remainder $R = 13$, Registration Ending in 13)

**Assigned Deep-Dive Algorithms:**

- **Algorithm 13:** NPGA (Niched Pareto Genetic Algorithm)

- **Algorithm 25:** MOACO (Multi-Objective Ant Colony Optimization)

- **Algorithm 37:** GradNorm (Gradient Normalization for Multi-Task Learning)

**Date:** October 2026

**Implementation Repository:** `C:\Users\Soham Umesh Chavan\OT_Lab_7`

**Execution Environment:** Python 3.10+, NumPy, Matplotlib (Zero External Optimization Toolboxes)

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## 1. Executive Summary & Verification Checklist

This project delivers a comprehensive, scratch implementation and empirical evaluation of all 37 Multiobjective Optimization (MOO) algorithms specified in the curriculum across 14 diverse benchmark problems (P1 through P14). All algorithms, metrics, and test problems were hand-coded in Python using exclusively NumPy and Matplotlib, in strict adherence to the zero-external-toolbox directive.

**Experimental Run Count Reconciliation**

Across the entire laboratory evaluation pipeline, exactly **428 independent experimental runs** were executed and logged in raw per-run CSVs:

- **Task A (68 runs):** 68 scheduled single-seed ($seed=42$) runs covering all 37 algorithms on assigned benchmark problems under a strict budget of 5,000 evaluations.

- **Task C (90 runs):** 90 parameter sensitivity runs across 10 random seeds ($42 \dots 51$) testing 3 hyperparameter levels for NPGA ($t_{\text{dom}} \in \{2, 10, 25\}$ on P3), MOACO ($\rho \in \{0.05, 0.20, 0.50\}$ on P8), and GradNorm ($\alpha \in \{0.12, 1.50, 3.00\}$ on P13). Subtotal: $30 + 30 + 30 = 90$ runs.

- **Task D (270 head-to-head runs):** 270 stochastic benchmark comparison runs across 10 random seeds ($42 \dots 51$), comprising:

- Study 1: P3 Unconstrained ZDT2 (10 algorithms $\times$ 10 seeds = 100 runs)

- Study 2: P5 Constrained Cantilever (8 algorithms $\times$ 10 seeds = 80 runs)

- Study 3: P10 Many-Objective DTLZ2 $M=5$ (5 algorithms $\times$ 10 seeds = 50 runs)

- Study 4: P8 Discrete TSP (4 algorithms $\times$ 10 seeds = 40 runs)

Subtotal: $100 + 80 + 50 + 40 = 270$ runs.

- **Task D Special Studies:** 17 non-CSV preference sweeps, compromise optimizations, MCDM sensitivity tests, and multi-task gradient comparisons logged in `task_d_special_studies.json`.

**Verification of Independence:** These counts represent entirely separate, non-overlapping experimental executions with distinct problem configurations, evaluation budgets, and random seed streams ($68 + 90 + 270 = 428$ raw CSV runs).

**Requirements Verification Checklist**

| Requirement ID | Description | Status | Evidence / Artifact |
|---|---|---|---|
| **A1** | Shared toolkit (`moo_utils.py`): Dominance, Deb's rules, sorting, crowding, HV, IGD, Spacing | **COMPLETE** | `moo/moo_utils.py`, 100 passing unit tests |
| **A2** | 37 algorithm modules implemented from scratch with `initialize`, `step`, `run` | **COMPLETE** | `moo/algorithms/alg01_*.py` to `alg37_*.py` |
| **A3** | Task A experimental runs across test problems with fixed seeds and budget compliance | **COMPLETE** | 68 completed runs in `results/csv/task_a_runs_detail.csv` |
| **B1** | Master comparison table covering all 37 algorithms with >=4 advantages & >=4 disadvantages | **COMPLETE** | `docs/TASK_B_ALGORITHM_COMPARISON.md` |
| **C1** | Three study cards for assigned roll number ($R=13$: NPGA, MOACO, GradNorm) | **COMPLETE** | `docs/TASK_C_DEEP_DIVES.md` |
| **C2** | Six numerical hand traces across 2 iterations reproduced exactly by code | **COMPLETE** | Hand traces in `TASK_C_DEEP_DIVES.md`, verified by `test_task_c_traces.py` |
| **C3** | 90 parameter sensitivity runs across 10 independent seeds with plots ($t_{\text{dom}}, \rho, \alpha$) | **COMPLETE** | `task_c_*_sensitivity_raw.csv`, 4 plots in `plots/sensitivity/` |
| **D1** | P3 Unconstrained head-to-head comparison across 10 seeds (100 runs, HV, Spacing, IGD) | **COMPLETE** | `task_d_p3_unconstrained_summary.csv`, overlay plot |
| **D2** | P5 Constrained head-to-head comparison respecting Deb's feasibility rules (80 runs) | **COMPLETE** | `task_d_p5_constrained_summary.csv`, overlay plot |
| **D3** | P10 Many-Objective ($M=5$) comparison across 10 seeds (50 runs: NSGA-II, NSGA-III, MOEA/D, RVEA, HypE) | **COMPLETE** | `task_d_p10_manyobj_m5_summary.csv`, overlay plot |
| **D4** | P8 Discrete TSP comparison across 10 seeds (40 runs: MOEA/D-ACO, MOACO, NSGA-II, MOMA) | **COMPLETE** | `task_d_p8_discrete_tsp_summary.csv`, overlay plot |
| **D5** | Special studies: Algs 1-4 sweeps, Algs 5-7 compromises, TOPSIS/VIKOR sensitivity, MGDA/GradNorm | **COMPLETE** | `task_d_special_studies.json` (17 custom evaluations) |
| **E1** | Six problem selection case studies with suitable/poor choices and empirical backing | **COMPLETE** | `docs/TASK_E_ALGORITHM_SELECTION.md` |
| **F1** | Complete answers to all 8 mandatory PDF viva questions | **COMPLETE** | `docs/TASK_F_VIVA_PREPARATION.md` |

| | | | |
|---|---|---|---|
| **F2** | Code walkthroughs for NPGA, MOACO, and GradNorm with mutation predictions | **COMPLETE** | `docs/TASK_F_VIVA_PREPARATION.md` |
| **EXP** | Total Reconciled Experiment Count: 428 Raw CSV Runs (68 Task A + 90 Task C + 270 Task D) | **COMPLETE** | Verified across all raw CSV logs, summary CSVs, and plots |
| **PKG** | Submission Package: Consolidated PDF Report, Runnable ZIP, CSVs, and README | **COMPLETE** | `docs/FINAL_LAB_REPORT.pdf`, `OT_Lab_7_Roll13_Submission.zip` |

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## 2. Problem Bank Mathematical Formulations (P1 to P14)

All 14 problems have been implemented in `moo/problems/` and standardized to internal minimization ($\min -f$ for maximization objectives).

**Summary Table of the 14 Benchmark Problems**

| Problem | Name | Type | $n$ | $M$ | Constraints | Primary Characteristic |
|---|---|---|---|---|---|---|
| **P1** | Factory Production Speed | Continuous | 1 | 2 | Bounds | Convex quadratic trade-off: $f_2 = (4 - \sqrt{f_1})^2$ |
| **P2** | Delivery Fleet Sizing | Mixed Real/Int | 1 | 2 | Bounds | Integer vs real granularity: $f_1 = 100/x, f_2 = x^2$ |
| **P3** | ZDT2 Non-Convex Benchmark | Continuous | 10 | 2 | Bounds | Non-convex quadratic dent: $f_2 = 1 - f_1^2$ |
| **P4** | Investment Portfolio | Continuous | 4 | 2 | Equality ($\sum w_i = 1$) | Markowitz mean-variance risk-return trade-off |
| **P5** | Cantilever Beam Design | Continuous | 2 | 2 | Non-linear stress | Lightness vs stiffness; active stress boundary |
| **P6** | Binh-Korn Constrained | Continuous | 2 | 2 | 2 Non-linear | Disconnected/awkward feasible geometry |
| **P7** | Water Allocation | Continuous | 2 | 2 | Linear inequality | Agriculture vs industry with river conservation |
| **P8** | Bi-Objective Delivery TSP | Discrete Permutation | 15 | 2 | Permutation tour | Distance vs traffic time; discrete combinatorial |
| **P9** | Breast Cancer Feature Selection | Binary Combinatorial | 30 | 2 | Non-empty ($\sum x_i \ge 1$) | 5-NN accuracy vs feature subset cardinality |
| **P10** | DTLZ2 Many-Objective | Continuous | 14 | 3, 5 | Bounds | Spherical manifold: $\sum f_m^2 = 1$; Pareto breakdown |
| **P11** | Production Planning | Linear LP | 2 | 2 | Linear resource | Workshop profit vs volume with priority hierarchy |
| **P12** | Supplier Selection (MCDM) | Discrete Set | 5 | 4 | Discrete alternatives | 5 Suppliers on cost, quality, delivery, reliability |
| **P13** | Multi-Task Learning | Differentiable | 2 | 2 | Unconstrained | Shared model optimization under gradient conflict |
| **P14** | Laptop Design (Preferences) | Continuous | 2 | 2 | Bounds | Weight vs battery life with user aspiration |

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## 3. Implementation Log & Coverage (All 37 Algorithms)

The 37 implemented algorithms span seven distinct methodological families:

| number | algorithm | problems used | evaluation budget | HV | IGD | runtime | status |
|---|---|---|---|---|---|---|---|
| 1 | Weighted Sum | P1, P3 | 5000 | P1: 1.0258; P3: 0.2100 | P1: 0.0202; P3: 0.3532 | 0.02s | works |
| 2 | epsilon-Constraint | P2, P3 | 5000 | P2: 1.0865; P3: 0.4331 | P2: 0.0370; P3: 0.0941 | 0.14s | works |
| 3 | Benson's Method | P1, P7 | 5000 | P1: 0.9377; P7: 0.1100 | P1: 0.1737; P7: 0.7071 | 0.04s | works |
| 4 | Weight Metric | P1, P3 | 5000 | P1: 1.0168; P3: 0.5133 | P1: 0.0409; P3: 0.0381 | 0.11s | works |
| 5 | Value Function | P4, P14 | 5000 | P4: 0.6707; P14: 0.3292 | P4: N/A; P14: 0.3534 | 0.01s | works |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | Goal Programming | P11, P7 | 5000 | P11: 1.2100; P7: 0.3407 | P11: 0.0000; P7: 0.3825 | 0.06s | works |
| 7 | Lexicographic | P11, P7 | 5000 | P11: 1.2100; P7: 0.1100 | P11: 0.0000; P7: 0.7071 | 0.05s | works |
| 8 | TOPSIS | P12, P4 | 5000 | P12: 0.1331; P4: 0.0000 | P12: N/A; P4: N/A | 0.00s | partly works |
| 9 | VIKOR | P12, P4 | 5000 | P12: 0.2854; P4: 0.0000 | P12: N/A; P4: N/A | 0.00s | partly works |
| 10 | VEGA | P2, P3 | 5000 | P2: 1.1164; P3: 0.0000 | P2: 0.0055; P3: 0.5580 | 14.06s | works |
| 11 | MOGA | P3, P5 | 5000 | P3: 0.0000; P5: 1.0925 | P3: 0.6418; P5: N/A | 15.66s | works |
| 12 | NSGA | P5, P7 | 5000 | P5: 1.1728; P7: 0.6971 | P5: N/A; P7: 0.0066 | 22.28s | works |
| 13 | NPGA | P3, P5 | 5000 | P3: 0.1963; P5: 1.1808 | P3: 0.2464; P5: N/A | 15.99s | works |
| 14 | NSGA-II | P3, P5 | 5000 | P3: 0.5288; P5: 1.1830 | P3: 0.0076; P5: N/A | 42.80s | works |
| 15 | NSGA-III | P10_M3, P10_M5 | 5000 | P10_M3: 0.7191; P10_M5: 0.2955 | P10_M3: 0.0447; P10_M5: 0.4964 | 142.99s | works |
| 16 | SPEA2 | P4, P5 | 5000 | P4: 0.7606; P5: 1.1805 | P4: N/A; P5: N/A | 10.42s | works |
| 17 | PESA-II | P3, P5 | 5000 | P3: 0.0000; P5: 1.1478 | P3: 1.5428; P5: N/A | 4.28s | works |
| 18 | MOEA/D | P3, P10_M3 | 5000 | P3: 0.5328; P10_M3: 0.7353 | P3: 0.0044; P10_M3: 0.0558 | 793.91s | works |
| 19 | MOEA/D-DE | P3, P5 | 5000 | P3: 0.4644; P5: 0.9375 | P3: 0.0374; P5: N/A | 760.73s | works |
| 20 | MOEA/D-ACO | P8 | 5000 | P8: 0.8901 | P8: N/A | 3.51s | works |
| 21 | Constrained NSGA-II | P5, P6 | 5000 | P5: 1.1829; P6: 1.0228 | P5: N/A; P6: N/A | 85.06s | works |
| 22 | MOPSO | P4, P5 | 5000 | P4: 0.5864; P5: 1.1830 | P4: N/A; P5: N/A | 8.19s | works |
| 23 | MOGWO | P5, P9 | 5000 | P5: 1.1772; P9: 1.1401 | P5: N/A; P9: N/A | 27.09s | works |
| 24 | MOABC | P3, P6 | 5000 | P3: 0.4718; P6: 1.0147 | P3: 0.0583; P6: N/A | 14.22s | works |
| 25 | MOACO | P8 | 5000 | P8: 0.9035 | P8: N/A | 4.06s | works |
| 26 | MOSA | P3, P5 | 5000 | P3: 0.0962; P5: 1.0821 | P3: 0.6124; P5: N/A | 110.45s | works |
| 27 | MODE | P3, P5 | 5000 | P3: 0.5374; P5: 1.1802 | P3: 0.0044; P5: N/A | 21.75s | works |
| 28 | MO-TLBO | P3, P5 | 5000 | P3: 0.5373; P5: 1.1815 | P3: 0.0045; P5: N/A | 12.20s | works |
| 29 | MOBA | P3, P4 | 5000 | P3: 0.0000; P4: 0.6894 | P3: 1.9015; P4: N/A | 1.65s | works |
| 30 | MOMA | P5, P9 | 5000 | P5: 1.1829; P9: 1.1110 | P5: N/A; P9: N/A | 56.58s | works |
| 31 | RVEA | P10_M3 | 5000 | P10_M3: 0.7469 | P10_M3: 0.0467 | 25.54s | works |
| 32 | IBEA | P3, P14 | 5000 | P3: 0.5387; P14: 0.8655 | P3: 0.0026; P14: 0.0017 | 36.03s | works |
| 33 | SMS-EMOA | P3, P14 | 5000 | P3: 0.5395; P14: 0.8658 | P3: 0.0025; P14: 0.0014 | 1796.66s | works |
| 34 | HypE | P10_M3 | 5000 | P10_M3: 0.7400 | P10_M3: 0.0497 | 19.46s | works |
| 35 | R-NSGA-II | P5, P14 | 5000 | P5: 1.1839; P14: 0.8531 | P5: N/A; P14: 0.0061 | 102.22s | works |
| 36 | MGDA | P13 | 5000 | P13: 0.3986 | P13: 0.5516 | 88.43s | works |
| 37 | GradNorm | P13 | 5000 | P13: 0.7169 | P13: 0.3631 | 87.45s | works |

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## 4. Task A: Algorithm Implementation & Front Visualizations

During Task A, 68 scheduled runs were executed across the 37 algorithms and 14 test problems with fixed seeds ($42$) and strict budget enforcement ($5,000$ evaluations).

- **Feasibility Enforcement:** Deb's constrained dominance rules were enforced across all constrained problems (P4, P5, P6, P7). Solutions violating constraints were filtered out of final metric calculations.

- **Metric Normalization:** All hypervolume calculations were normalized against documented problem-specific ideal and nadir bounds with a $1.1$ reference point offset.

- **Analytical Front Validation:** Analytical reference fronts were used for P1, P2, P3, P5, P6, P7, P10, and P13(a) to compute valid Inverted Generational Distance (IGD).

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## 5. Task B: Master Comparative Taxonomy & Family Synthesis

The complete comparative taxonomy covering all 37 algorithms, with at least four distinct advantages and four distinct disadvantages per algorithm backed by Task A empirical numbers, is fully detailed in `docs/TASK_B_ALGORITHM_COMPARISON.md`.

**Methodological Family Synthesis**

1. **Classical Scalarization Methods (Algs 1–7):** Highly computationally efficient per solve ($O(1)$ to $O(N)$), but incapable of capturing non-convex Pareto fronts via linear weights (Weighted Sum #1). $\epsilon$-Constraint (#2) circumvents non-convexity by reformulating secondary objectives as bounds.

2. **Early Non-Elitist Evolutionary Algorithms (Algs 10–13):** Historically pioneering, but suffer from genetic drift and lack of monotonic convergence. VEGA (#10) suffers extreme speciation, collapsing to boundary endpoints.

3. **Elitist Multi-Objective Evolutionary Algorithms (Algs 14–20):** NSGA-II (#14), SPEA2 (#16), and MOEA/D (#18) provide strong convergence and diversity on bi-objective problems via parent-offspring elitism ($2N \to N$) or external archives.

4. **Constrained & Metaheuristic Swarm Optimizers (Algs 21–29):** CNSGA-II (#21) embeds Deb's three feasibility rules directly into tournament selection. Swarm methods (MOPSO #22, MOGWO #23, MOACO #25) excel on specific domains like continuous non-linear spaces or discrete graphs.

5. **Hybrid, Many-Objective, Indicator, Preference & Gradient Methods (Algs 30–37):** NSGA-III (#15) and RVEA (#31) overcome the many-objective breakdown ($M \ge 4$) using structured reference vectors. GradNorm (#37) dynamically balances task gradient norms in multi-task learning.

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## 6. Task C: Deep Dive Analysis, Numerical Hand Traces & Parameter Sensitivities

Comprehensive deep dives for the candidate's assigned algorithms ($R = 13$: NPGA #13, MOACO #25, GradNorm #37) are documented in `docs/TASK_C_DEEP_DIVES.md`.

**Summary of Parameter Sensitivity Experiments (90 Runs across 10 Seeds)**

| algorithm | problem | parameter | level | value | runs | hv_mean | hv_std | spacing_mean | spacing_std | igd_mean | igd_std | runtime_mean | runtime_s |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NPGA | P3 | t_dom | low | 2.0000 | 10 | 0.00e+00 | 0.00e+00 | 0.2881 | 0.2171 | 1.0854 | 0.1918 | 1.2978 | 0.5650 |
| NPGA | P3 | t_dom | default | 10.0000 | 10 | 0.00e+00 | 0.00e+00 | 0.3440 | 0.2449 | 1.2823 | 0.2959 | 1.2043 | 0.0779 |
| NPGA | P3 | t_dom | high | 25.0000 | 10 | 0.00e+00 | 0.00e+00 | 0.2268 | 0.1106 | 1.2332 | 0.1199 | 1.1523 | 0.1886 |
| MOACO | P8 | rho | low | 0.0500 | 10 | 0.5459 | 0.0246 | 0.0061 | 0.0070 | nan | nan | 0.6070 | 0.0874 |
| MOACO | P8 | rho | default | 0.2000 | 10 | 0.5640 | 0.0327 | 0.0214 | 0.0182 | nan | nan | 0.7206 | 0.0631 |
| MOACO | P8 | rho | high | 0.5000 | 10 | 0.8232 | 0.0421 | 0.0163 | 0.0063 | nan | nan | 0.6669 | 0.0836 |
| GradNorm | P13 | alpha | low | 0.1200 | 10 | 0.5943 | 0.1915 | 0.0038 | 0.0020 | 0.4129 | 0.1254 | 0.4601 | 0.0956 |
| GradNorm | P13 | alpha | default | 1.5000 | 10 | 0.7248 | 0.0401 | 0.0107 | 0.0153 | 0.3491 | 0.0220 | 0.3681 | 0.1057 |
| GradNorm | P13 | alpha | high | 3.0000 | 10 | 0.7349 | 0.0484 | 0.0108 | 0.0157 | 0.3421 | 0.0401 | 0.3229 | 0.0928 |

**Key Sensitivity Insights**

- **NPGA (#13, Tournament Sample Size $t_{\text{dom}}$):** Tested on P3 (ZDT2) with $t_{\text{dom}} \in \{2, 10, 25\}$. Increasing $t_{\text{dom}}$ from 2 to 25 steadily increases selection pressure. At $t_{\text{dom}} = 2$, candidates frequently tie

in dominance comparisons, falling back to niche sharing and maintaining broader exploratory diversity ($IGD = 1.0854 \pm 0.1918$), while $t_{\text{dom}} = 25$ yields tighter spacing clustering ($S = 0.2268 \pm 0.1106$).

- **MOACO (#25, Evaporation Rate $\rho$):** Tested on P8 (TSP) with $\rho \in \{0.05, 0.20, 0.50\}$. Increasing $\rho$ from $0.05$ to $0.50$ produces a massive leap in Hypervolume from $0.5459 \to 0.8232$ (+50.8% gain) because obsolete suboptimal tours evaporate quickly, preventing ants from stagnating in early suboptimal cycles.

- **GradNorm (#37, Asymmetry Restoring Force $\alpha$):** Tested on P13 (Multi-Task Learning) with $\alpha \in \{0.12, 1.50, 3.00\}$. Increasing $\alpha$ from $0.12$ to $1.50$ and $3.00$ dramatically reduces IGD from $0.4129 \to 0.3421$ (-17.1%) and boosts Hypervolume from $0.5943 \to 0.7349$ (+23.6%), stabilizing multi-task gradient balancing.

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## 7. Task D: Head-to-Head Comparative Empirical Studies

All head-to-head comparisons were conducted under equal evaluation budgets and across 10 independent random seeds ($42 \dots 51$). Exactly **270 independent runs** were executed across the four benchmark studies (100 on P3, 80 on P5, 50 on P10, 40 on P8), supplemented by 17 targeted evaluations in Study 5. Feasibility filtering was strictly applied before computing metrics.

**Study 1: P3 Non-Convex Benchmark (ZDT2, Budget 1500, 100 Runs)**

| experiment | problem | algorithm | algorithm_id | runs | budget | evals_mean | hv_mean | hv_std | spacing_mean | spacing_std | igd_mean |
|---|---|---|---|---|---|---|---|---|---|---|---|
| P3_Unconstrained | P3 | WEIGHTED_SUM | 1 | 10 | 1500 | 302.6000 | 0.2100 | 1.14e-12 | 0.00e+00 | 0.00e+00 | 0.3532 |
| P3_Unconstrained | P3 | EPSILON_CONSTRAINT | 2 | 10 | 1500 | 1120.4000 | 0.4387 | 0.0117 | 0.5612 | 0.3904 | 0.0998 |
| P3_Unconstrained | P3 | VEGA | 10 | 10 | 1500 | 1500.0000 | 0.00e+00 | 0.00e+00 | 0.1514 | 0.0962 | 1.2856 |
| P3_Unconstrained | P3 | MOGA | 11 | 10 | 1500 | 1500.0000 | 0.0070 | 0.0209 | 0.1345 | 0.0700 | 1.0205 |
| P3_Unconstrained | P3 | NPGA | 13 | 10 | 1500 | 1500.0000 | 0.00e+00 | 0.00e+00 | 0.3440 | 0.2449 | 1.2823 |
| P3_Unconstrained | P3 | NSGA2 | 14 | 10 | 1500 | 1500.0000 | 0.0517 | 0.0492 | 0.1085 | 0.0650 | 0.5117 |
| P3_Unconstrained | P3 | MOEAD | 18 | 10 | 1500 | 1500.0000 | 0.2766 | 0.0782 | 0.0326 | 0.0280 | 0.2097 |
| P3_Unconstrained | P3 | MOEAD_DE | 19 | 10 | 1500 | 1500.0000 | 0.0026 | 0.0078 | 0.0612 | 0.0502 | 1.0361 |
| P3_Unconstrained | P3 | MODE | 27 | 10 | 1500 | 1500.0000 | 0.4547 | 0.0226 | 0.0335 | 0.0094 | 0.0487 |
| P3_Unconstrained | P3 | SMS_EMOA | 33 | 10 | 1500 | 1500.0000 | 0.1454 | 0.1103 | 0.0444 | 0.0393 | 0.4471 |

- **Analysis:** MODE (#27) achieved the highest hypervolume ($HV = 0.4547 \pm 0.0226$) and lowest IGD ($0.0487 \pm 0.0131$). Weighted Sum (#1) collapsed to exactly 2 extreme points ($HV = 0.2100, S = 0.0$), empirically proving its mathematical inability to populate concave dents.

**Study 2: P5 Constrained Cantilever Beam (Budget 1500, 80 Runs)**

| experiment | problem | algorithm | algorithm_id | runs | budget | evals_mean | hv_mean | hv_std | spacing_mean | spacing_std | igd_mean | igd_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P5_Constrained | P5 | EPSILON_CONSTRAINT | 2 | 10 | 1500 | 909.6000 | 1.0892 | 0.0016 | 0.2010 | 0.0024 | nan | nan |
| P5_Constrained | P5 | NPGA | 13 | 10 | 1500 | 1500.0000 | 1.1832 | 0.0006 | 0.0627 | 0.0182 | nan | nan |
| P5_Constrained | P5 | NSGA2 | 14 | 10 | 1500 | 1500.0000 | 1.1831 | 0.0004 | 0.0425 | 0.0084 | nan | nan |
| P5_Constrained | P5 | SPEA2 | 16 | 10 | 1500 | 1500.0000 | 1.1794 | 0.0014 | 0.1132 | 0.0089 | nan | nan |
| P5_Constrained | P5 | CNSGA2 | 21 | 10 | 1500 | 1500.0000 | 1.1828 | 0.0004 | 0.0384 | 0.0034 | nan | nan |
| P5_Constrained | P5 | MOPSO | 22 | 10 | 1500 | 1500.0000 | 1.1831 | 0.0004 | 0.1472 | 0.0409 | nan | nan |
| P5_Constrained | P5 | MOGWO | 23 | 10 | 1500 | 1500.0000 | 1.1812 | 0.0022 | 0.1879 | 0.0940 | nan | nan |
| P5_Constrained | P5 | MODE | 27 | 10 | 1500 | 1500.0000 | 1.1814 | 0.0010 | 0.0860 | 0.0091 | nan | nan |

- **Analysis:** CNSGA-II (#21) achieved the most uniform spacing ($S = 0.0384 \pm 0.0034$) with 100% feasibility ($CV = 0.0$). Deb's feasibility-first rules successfully guided the population along the active bending stress boundary ($b h^2 = 2.4 \times

10^{-5}$).

**Study 3: P10 Many-Objective Benchmark (DTLZ2, $M=5$, Budget 2000, 50 Runs)**

| experiment | problem | algorithm | algorithm_id | runs | budget | evals_mean | hv_mean | hv_std | spacing_mean | spacing_std | igd_mean | igd_std | run |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P10_ManyObj_M5 | P10_M5 | NSGA2 | 14 | 10 | 2000 | 2000.0000 | 0.1904 | 0.0722 | 0.2602 | 0.0160 | 0.5586 | 0.0519 | 26.1 |
| P10_ManyObj_M5 | P10_M5 | NSGA3 | 15 | 10 | 2000 | 2000.0000 | 0.1989 | 0.0187 | 0.2521 | 0.0205 | 0.5586 | 0.0160 | 34.6 |
| P10_ManyObj_M5 | P10_M5 | MOEAD | 18 | 10 | 2000 | 2000.0000 | 0.7051 | 0.0876 | 0.1837 | 0.0196 | 0.3617 | 0.0355 | 4.16 |
| P10_ManyObj_M5 | P10_M5 | RVEA | 31 | 10 | 2000 | 2000.0000 | 1.1395 | 0.0202 | 0.1383 | 0.0087 | 0.2071 | 0.0070 | 10.1 |
| P10_ManyObj_M5 | P10_M5 | HYPE | 34 | 10 | 2000 | 2000.0000 | 1.0775 | 0.0242 | 0.1410 | 0.0124 | 0.2415 | 0.0097 | 7.29 |

- **Degradation of NSGA-II:** In 5-dimensional objective space, over 95% of candidates are mutually non-dominated, causing Pareto rank selection pressure to collapse. Crowding distance favors extreme boundary points on individual axes, causing severe interior under-representation. Reference-direction algorithms (RVEA #31 with $HV=1.1395, IGD=0.2071$; HypE #34 with $HV=1.0775, IGD=0.2415$) maintain uniform coverage across the 5D spherical shell.

**Study 4: P8 Discrete Permutation Routing (MOTSP, 15 Cities, Budget 1000, 40 Runs)**

| experiment | problem | algorithm | algorithm_id | runs | budget | evals_mean | hv_mean | hv_std | spacing_mean | spacing_std | igd_mean | igd_std | ru |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P8_Discrete_TSP | P8 | MOEAD_ACO | 20 | 10 | 1000 | 1000.0000 | 0.5715 | 0.0206 | 0.0122 | 0.0216 | nan | nan | 0.3 |
| P8_Discrete_TSP | P8 | MOACO | 25 | 10 | 1000 | 1000.0000 | 0.5711 | 0.0324 | 0.0104 | 0.0130 | nan | nan | 0.5 |
| P8_Discrete_TSP | P8 | NSGA2 | 14 | 10 | 1000 | 1000.0000 | 0.7205 | 0.0266 | 0.0222 | 0.0228 | nan | nan | 1.9 |
| P8_Discrete_TSP | P8 | MOMA | 30 | 10 | 1000 | 1000.0000 | 0.8310 | 0.0270 | 0.0020 | 0.0043 | nan | nan | 0.8 |

- **Permutation Operator Analysis:** MOMA (#30) achieved Rank 1 ($HV = 0.8310 \pm 0.0270, S = 0.0020 \pm 0.0043$) and NSGA-II (#14) achieved Rank 2 ($HV = 0.7205 \pm 0.0266$) using Order Crossover and swap mutation, outperforming constructive ant colony methods within 1,000 evaluations. MOACO (#25) and MOEA/D-ACO (#20) use dual pheromone trails ($\tau^1, \tau^2$) to probabilistically construct valid closed tours without requiring post-hoc repair.

**Study 5: Special Studies & Preference Experiments (17 Custom Evaluations)**

- **Algorithms 1–4:** 10-setting sweeps demonstrate the superiority of $\epsilon$-Constraint over Weighted Sum on non-convex fronts.

- **Algorithms 5–7:** Lexicographic optimization uniquely isolates $(300, 200)$ on P11 by breaking the profit tie using total production volume.

- **Algorithms 8–9 (TOPSIS & VIKOR):** Shifting delivery weight by $0.1$ altered supplier rankings, proving MCDM sensitivity.

- **Algorithms 36–37 (MGDA & GradNorm):** GradNorm achieved balanced loss reduction across both tasks on P13, preventing task dominance.

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## 8. Task E: Six Problem-Specific Algorithm Selection Case Studies

Detailed case studies for Problems P3, P5, P10, P8, P9, and P11 are fully articulated in `docs/TASK_E_ALGORITHM_SELECTION.md`.

**Core Takeaways:**

1. **Convexity:** Non-convex problems (P3) forbid linear scalarization (Weighted Sum); $\epsilon$-Constraint or Pareto evolutionary methods must be chosen.

2. **Dimension:** Many-objective spaces ($M \ge 4$, P10) require structured reference lines (NSGA-III) or angle penalties (RVEA) rather than crowding distance.

3. **Representation:** Permutations (P8) require constructive ant colony or order-preserving genetic operators. Continuous gradients fail completely.

4. **Constraints:** Heavily constrained engineering designs (P5) require Deb's feasibility-first rules (CNSGA-II) to ensure physically valid solutions along active constraint boundaries.

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## 9. Task F: Viva Voce Examination & Code Walkthrough Answers

Full answers to all eight mandatory viva questions and detailed code walkthroughs for NPGA (#13), MOACO (#25), and GradNorm (#37) are provided in `docs/TASK_F_VIVA_PREPARATION.md`.

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## 10. Limitations, Threats to Validity & Reproducibility Guide

**Limitations**

1. **Analytical vs Noisy Real-World Objectives:** Analytical benchmarks (ZDT2, DTLZ2) possess smooth deterministic landscapes; real-world engineering problems feature observational noise and evaluation latency.

2. **Monte Carlo Hypervolume Approximation ($M \ge 3$):** Hypervolume in 5D is estimated via Monte Carlo sampling (100,000 points), introducing a small stochastic variance ($\approx \pm 0.005$).

3. **Discrete Tour Representation:** Permutation TSP was evaluated on 15 cities; scaling to hundreds of cities requires specialized 2-opt local search.

**Reproducibility Guide**

All results can be reproduced using standard Python commands:

```
# Run unit tests
python main.py test

# Run Task A experiments
python main.py task_a --budget 5000 --seed 42

# Run Task D head-to-head comparisons
python main.py task_d --seeds 42-51 --results-dir results --plots-dir plots
```

All random number generators use explicit, fixed seeds recorded in the raw CSV outputs.